

Introduction to Ontology Engineering (Part 2)

A Two-Week Module for Artificial Intelligence Capstone

Chun-yien Chang, PhD Candidate

Department of Computer Science, NYCU

ccy@hptp.org

May 2026

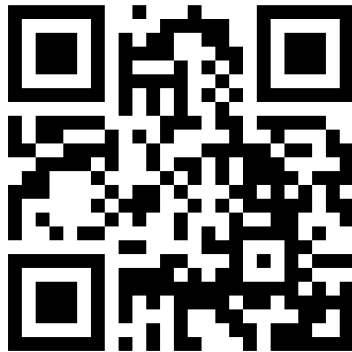
Before We Begin

A quick interaction check with Vevox

- ▶ Please join the Vevox session before we begin.
- ▶ We will use it for short checkpoints and concept-review questions.
- ▶ Your responses help me see which ideas need further clarification.

Registration format

- ▶ **First name:** Student ID
- ▶ **Last name:** Your name



Scan to join Vevox.

A Brief Recap

What we established in Part 1

Ontology as engineering

- ▶ An ontology makes a domain understanding explicit and structured.
- ▶ It addresses hidden assumptions and unstable meaning in data and systems.
- ▶ We began with a simple Student table to expose this problem.

The first technical steps

- ▶ RDF represents meaning through graph-like triples.
- ▶ Turtle gives us a readable textual form for RDF.
- ▶ Namespaces and URIs connect local statements to reusable Web vocabularies.

Where we ended

We moved from a local data table to a Web-oriented way of identifying, structuring, and connecting meaning.

From the World Wide Web to the Semantic Web

Why ontology engineering has deep roots in the Web

The Web Began as a Web of Documents

Tim Berners-Lee's original problem

The 1989 proposal

- ▶ Tim Berners-Lee proposed the World Wide Web at CERN.
- ▶ The problem was information management across many documents and systems.
- ▶ The solution centered on hypertext documents connected by links.

Core idea

A document can point to another document, and a browser can let users move across that network of links.

Starting point

The Web first became powerful by linking *documents*.

The First Website Is Still Accessible

A living artifact from the early Web

CERN preserves the first website

- ▶ The earliest Web page introduced the World Wide Web project itself.
- ▶ It described a system for giving broad access to linked documents.
- ▶ Remarkably, this historical page can still be visited today.

Quick browser exercise. Open the first website and inspect its source code.

1. Visit: `info.cern.ch` to browse the first website
2. Open *View Page Source* or browser developer tools. Notice how simple the early Web page is.

Why this matters

Before the Semantic Web aimed to make meaning machine-processable, the World Wide Web began with a remarkably simple document-linking structure.

Primary source: CERN. (n.d.-b). The world wide web project. <https://info.cern.ch/hypertext/WWW/TheProject.html>

After viewing the first website, what is the main structure you observed?

- A. A database of formal ontology classes.
- B. A simple document structure connected by hyperlinks.
- C. A JSON-LD block using schema.org vocabulary.
- D. An AI agent reasoning over user preferences.

Purpose of this question

This checks the starting point of the Web: linked documents before explicitly structured meaning.

This slide is intentionally hidden.

The content is reserved for the in-class session.

The Semantic Web Vision

An extension of the existing Web

“The Semantic Web is not a separate Web but an extension of the current one.”

The proposed shift

The existing Web links documents for human use. The Semantic Web aims to add well-defined meaning so that computers can more effectively process, combine, and act on information.

From linked documents → linked meaning

Primary source: Berners-Lee, T., Hendler, J., & Lassila, O. (2001). The semantic web. *Scientific American*, 284(5), 34–43.
<https://doi.org/10.1038/scientificamerican0501-34>

A Landmark Article

The Semantic Web vision in 2001

Tim Berners-Lee, James Hendler, and Ora Lassila's *The Semantic Web* presented the broader vision to a general scientific audience.



Primary source: Berners-Lee, T., Hendler, J., & Lassila, O. (2001). The semantic web. *Scientific American*, 284(5), 34–43.
<https://doi.org/10.1038/scientificamerican0501-34>

A 2001 Glossary

How the Semantic Web article framed its core terms

Glossary

HTML: Hypertext Markup Language. The language used to encode formatting, links and other features on Web pages. Uses standardized "tags" such as <H1> and <BODY> whose meaning and interpretation is set universally by the World Wide Web Consortium.

XML: eXtensible Markup Language. A markup language like HTML that lets individuals define and use their own tags. XML has no built-in mechanism to convey the meaning of the user's new tags to other users.

RESOURCE: Web jargon for any entity. Includes Web pages, parts of a Web page, devices, people and more.

URL: Uniform Resource Locator. The familiar codes (such as <http://www.sciam.com/index.html>) that are used in hyperlinks.

URI: Universal Resource Identifier. URLs are the most familiar type of URI. A URI defines or specifies an entity, not necessarily by naming its location on the Web.

RDF: Resource Description Framework. A scheme for defining information on the Web. RDF provides the technology for expressing the meaning of terms and concepts in a form that computers can readily process. RDF can use XML for its syntax and URIs to specify entities, concepts, properties and relations.

ONTOLOGIES: Collections of statements written in a language such as RDF that define the relations between concepts and specify logical rules for reasoning about them. Computers will "understand" the meaning of semantic data on a Web page by following links to specified ontologies.

AGENT: A piece of software that runs without direct human control or constant supervision to accomplish goals provided by a user. Agents typically collect, filter and process information found on the Web, sometimes with the help of other agents.

SERVICE DISCOVERY: The process of locating an agent or automated Web-based service that will perform a required function. Semantics will enable agents to describe to one another precisely what function they carry out and what input data are needed.

What Did the Semantic Web Need?

RDF, ontology languages, and agents

RDF

A graph-based way to state machine-readable relations among resources.

Ontologies

Shared vocabularies and formal structures for defining domain meaning.

Agents

Software systems that can use explicit meaning to combine information and support action.

The 2001 Semantic Web vision discussed RDF, ontologies, and intelligent agents. OWL was standardized shortly afterward as the W3C ontology language for representing richer knowledge on the Web.

The Agent Vision Was Already There

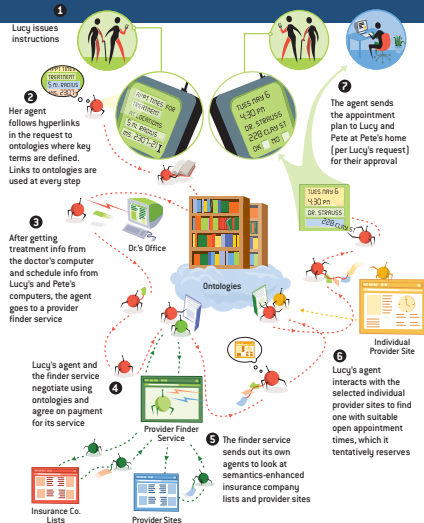
A striking part of the 2001 Semantic Web article

Already in 2001

The Semantic Web article did not only discuss RDF and ontologies. It also imagined *software agents* using semantic content on the Web to coordinate complex tasks.

In this example, an agent follows semantic links, interacts with services, and helps arrange a medical appointment under multiple constraints.

Primary source: Berners-Lee, T., Hendler, J., & Lassila, O. (2001). The semantic web. *Scientific American*, 284(5), 34–43.
<https://doi.org/10.1038/scientificamerican0501-34>



A Very Short Historical Sequence

Ontology engineering inherits a Web-scale ambition

Year	Milestone	Why it matters here
1989	WWW proposal	A Web of linked hypertext documents is proposed.
1999	RDF Recommendation	The Web gains a standard model for graph-based semantic statements.
2001	<i>The Semantic Web</i>	Berners-Lee, Hendler, and Lassila articulate the broader vision.
2004	OWL Recommendation	The Web gains a standard ontology language for richer formal meaning.

Selected standards referenced in this timeline:

- ▶ World Wide Web Consortium. (1999). Resource Description Framework (RDF) Model and Syntax Specification [W3C Recommendation, 22 February 1999]. <https://www.w3.org/TR/1999/REC-rdf-syntax-19990222/>
- ▶ World Wide Web Consortium. (2004). OWL Web Ontology Language Reference [W3C Recommendation, 10 February 2004]. <https://www.w3.org/TR/owl-ref/>

Ontology Is Actually Everywhere

Schema.org, JSON-LD, and structured meaning on the Web

What it provides

- ▶ A shared vocabulary for describing Web content.
- ▶ Types, properties, and relations that can be reused across websites.
- ▶ A practical bridge between local page content and machine-readable meaning.

Primary source: [Schema.org](https://schema.org/). (n.d.-c). Schema.org. <https://schema.org/>

Why it matters here

In Part 1, we used terms such as `schema:Person` and `schema:affiliation`.

These were not invented classroom labels: they come from a vocabulary already used at Web scale.

Where Did Schema.org Come From?

A shared vocabulary backed by major Web platforms

The founding coalition

- ▶ Google
- ▶ Microsoft
- ▶ Yahoo
- ▶ Yandex

Why this is significant

Schema.org was not created as a small academic vocabulary. It emerged from the need for a shared semantic vocabulary that Web publishers and large platforms could both use.

Implication

Semantic Web ideas did not remain only in standards documents. They became part of the practical infrastructure of large-scale Web interpretation.

From a Schema.org Entry to JSON-LD

The Product page shows how semantic markup is written

Observe the official example

- ▶ Visit: schema.org/Product
- ▶ Scroll to the examples.
- ▶ Open the **JSON-LD** tab.
- ▶ Notice that the example describes:
 - ▶ a battery replacement,
 - ▶ a *Virtual Pump Twin*,
 - ▶ a software module.

What appears in the markup?

- ▶ @context: connects the data to shared vocabularies.
- ▶ @graph: groups several described resources together.
- ▶ @type: states what kind of resource each item is.

This is where schema.org becomes visible inside ordinary Web markup: the vocabulary is written into a machine-readable JSON-LD block.

The Official JSON-LD Example

A graph embedded inside Web markup

```
<script type="application/ld+json">
{
  "@context": [
    "https://schema.org",
    ...
  ],
  "@graph": [
    {
      "@id": "urn:example:product:battery-x001",
      "@type": "Product",
      ...
    }
  ]
}
</script>
```

Two key clues

- ▶ @context connects the JSON block to shared vocabularies.
- ▶ @graph allows several related resources to be described together.

This is not ordinary JSON alone. It is JSON used to express structured semantic descriptions.

A Real Webpage with JSON-LD

Structured data inside a New Yorker article

Browser exercise

- ▶ Open the New Yorker article on Tim Berners-Lee.
- ▶ Use *View Page Source*.
- ▶ Search for:
 - ▶ `application/ld+json`
 - ▶ `NewsArticle`
 - ▶ `BreadcrumbList`

Why this example?

This is a real media webpage, not a tutorial page.

It uses schema.org vocabulary to expose machine-readable meaning in the page source.

The irony is useful: an article about Tim Berners-Lee itself contains Semantic-Web-style structured metadata.

Primary source: [Lucas, J. \(2025\). Tim berners-lee invented the world wide web. now he wants to save it. *The New Yorker*.](https://www.newyorker.com/magazine/2025/10/06/tim-berners-lee-invented-the-world-wide-web-now-he-wants-to-save-it)
<https://www.newyorker.com/magazine/2025/10/06/tim-berners-lee-invented-the-world-wide-web-now-he-wants-to-save-it>

The Page Identifies Itself as a News Article

Schema.org vocabulary in JSON-LD

```
<script type="application/ld+json">
{
  "@context": "http://schema.org",
  "@type": "NewsArticle",
  "headline":
    "Tim Berners-Lee Invented the
    World Wide Web. Now He Wants to Save It",
  "author": [{
    "@type": "Person",
    "name": "Julian Lucas"
  }],
  "datePublished":
    "2025-09-29T06:00:00.000-04:00"
}
</script>
```

What becomes explicit?

- ▶ The page describes a NewsArticle.
- ▶ The author is a Person.
- ▶ The article has a headline and publication date.

The visible article is written for readers. The JSON-LD block adds a structured description for machines.

The Markup Describes More Than the Article

Publisher, webpage, and navigation

```
"publisher": {
  "@type": "Organization",
  "name": "The New Yorker"
},
"mainEntityOfPage": {
  "@type": "WebPage",
  "@id": "https://www.newyorker.com/..."
}
...
{
  "@type": "BreadcrumbList",
  "itemListElement": [
    { "@type": "ListItem", "position": 1,
      "name": "Magazine" }
  ]
}
```

Semantic structure

- ▶ The publisher is an Organization.
- ▶ The article is tied to a specific WebPage.
- ▶ The page's breadcrumb path is also semantically described.

Structured data can describe content, provenance, publication context, and navigation structure.

Why Does Google Care?

Structured data as explicit clues about page meaning

Google's basic claim

- ▶ Structured data gives Google explicit clues about the meaning of a page.
- ▶ It helps classify page content.
- ▶ It can support richer search appearances.

Implementation guidance

- ▶ Most Google Search structured data uses schema.org vocabulary.
- ▶ Google supports JSON-LD, Microdata, and RDFa.
- ▶ JSON-LD is generally recommended when feasible.

Interpretation

From an ontology-engineering perspective, structured data turns part of a webpage's intended meaning into an explicit, reusable semantic artifact.

Primary source: [Google Search Central](https://developers.google.com/search/docs/appearance/structured-data/intro-structured-data). (n.d.). Introduction to structured data markup in google search.
<https://developers.google.com/search/docs/appearance/structured-data/intro-structured-data>

Which schema.org type did the New Yorker JSON-LD use to describe the article itself?

- A. Organization
- B. NewsArticle
- C. BreadcrumbList
- D. ImageObject

Purpose of this question

This checks whether you can read the content of a JSON-LD block, not only recognize that structured data exists.

This slide is intentionally hidden.

The content is reserved for the in-class session.

Google: Structured data for web developers

<https://youtu.be/hUHjeDy1hE8>

Viewing focus

As you watch, focus on how structured data gives Web pages a shared layer of meaning. The key issue is not only search ranking or display format, but how a large Web environment can interpret page content through common vocabularies such as schema.org.

What is the main ontology-engineering idea behind structured data on the Web?

- A. It makes webpages visually more attractive.
- B. It replaces HTML with a different programming language.
- C. It gives page content explicit meaning through shared vocabularies.
- D. It guarantees that every search result will rank higher.

Purpose of this question

This connects Google structured data back to ontology engineering: shared vocabulary, explicit meaning, and large-scale interoperability.

This slide is intentionally hidden.

The content is reserved for the in-class session.

A Practical Semantic Web Outcome

Schema.org connects ontology vocabulary to everyday search

**A webpage can be readable by humans
and semantically informative to machines.**

Vocabulary	Encoding	Use
Schema.org	JSON-LD	Search interpretation

Transition

This is one reason ontology is already everywhere: it is embedded in the infrastructure by which the Web is interpreted, indexed, and presented back to us.

Primary source: Google Search Central. (n.d.). Introduction to structured data markup in google search.
<https://developers.google.com/search/docs/appearance/structured-data/intro-structured-data>

Robotics-Centric Ontologies

A small map before independent study

Ontology	Main focus	Why it matters for robotics concepts
CORA / IEEE 1872	Core concepts for robotics and automation	Provides a standardized upper-level robotics vocabulary for representing and communicating domain knowledge.
KnowRob	Knowledge processing for robots	Combines symbolic knowledge, reasoning, and grounded robot knowledge for task-level understanding.
SOMA	Socio-physical model of activities	Models everyday activities, physical context, social context, roles, and action-related knowledge.
ORO	OpenRobots Ontology	Supports cognitive robotic applications through OWL/RDF knowledge representation and reasoning.

These examples show that robotics ontology is not only about naming robot parts. It also concerns actions, tasks, environments, affordances, activities, and shared conceptual commitments.

Sources: IEEE. (2015). Ieee 1872-2015: Ieee standard ontologies for robotics and automation [Core Ontology for Robotics and Automation (CORA)]. <https://standards.ieee.org/standard/1872-2015.html> KnowRob. (n.d.). Knowrob: Knowledge processing for robots. <https://knowrob.org/> Beßler, D., Porzel, R., Pomarlan, M., Vyas, A., Höffner, S., Beetz, M., Malaka, R., & Bateman, J. (2021). Foundations of the socio-physical model of activities (soma) for autonomous robotic agents. *Proceedings of the 2021 International Conference on Formal Ontology in Information Systems*. <https://arxiv.org/abs/2011.11972> Lemaignan, S., Ros, R., Mösenlechner, L., Alami, R., & Beetz, M. (2010). Oro, a knowledge management platform for cognitive architectures in robotics. *Proceedings of the 2010 IEEE/RSJ International Conference on Intelligent Robots and Systems*.

Independent Study Opportunity

Knowledge representation and foundational ontology for Physical AI

- ▶ Focus: ontology engineering beyond this short module.
- ▶ Context: AI, robotics, semantic interoperability, and structured knowledge.
- ▶ Format: guided reading, concept modeling, and small ontology artifacts as outcome.

This is suitable for those who want to continue from course exposure to research-oriented ontology engineering practice.

Knowledge Representation and Formal Ontology for Physical AI

Contact: Chun-yien Chang (ccy@hptp.org)

Description

As artificial intelligence transitions from digital environments to the physical world, the deployment of “Physical AI” faces critical challenges regarding safety, interoperability, and explainability. While modern learning-based models excel at perception, they often operate as opaque black boxes, lacking the formal semantics required for verifiable reasoning in dynamic, real-world settings.

This research explores the reintegration of structured knowledge—specifically through upper and middle ontologies—into modern robotic architectures [1]. By investigating how foundational ontologies provide the essential semantic grounding for cross-domain interoperability [2], the study focuses on the theoretical underpinnings of ontology-augmented verification. This semantic integration enables robots to formally reason about their environment, contrastively evaluate competing task plans, and narrate their decision-making processes transparently [3]. Furthermore, this research investigates how semantic modeling enhances advanced systems engineering practices, such as automated Failure Mode and Effects Analysis (FMEA), to ensure trustworthy and beneficial AI operations [4].

Selected References

- [1] C.-y. Chang, Y. Chen, and Y.-p. Chen, “Reinvigorating structured knowledge and ontologies for trustworthy and beneficial ai and robotics,” in *Formal Ontology in Information Systems - Proceedings of the 15th International Conference, FOIS 2025*, 2025, pp. 125–136.
- [2] E. Aguado, A. Pease, R. Sanz, C. Hernández, J. D. Peña Narváez, G. Rezende, and A. Gabriel, “D1.2 Analysis of Ontologies and of Ontological Formalisms - The CORESENSE Ontological Approach,” The CORESENSE Consortium, Tech. Rep. CS-046-v-1.1, 2023.
- [3] A. Olivares-Alarcos, S. Foix, J. Borràs, G. Canal, and G. Alenyà, “Ontological foundations for contrastive explanatory narration of robot plans,” *Information Sciences*, 2026.
- [4] H. Younis, S. Kabir, F. Campean, P. Bonnaud, and D. Delaux, “AI- and Ontology-Based Enhancements to FMEA for Advanced Systems Engineering: Current Developments and Future Directions,” *Applied Sciences*, vol. 16, no. 5, 2026.